

Using program-level completion metrics to support inclusive postgraduate cultures

Nicholas T. Young¹, Christopher Overton¹, Bill Bridges², James T. Lavery²

¹ Department of Physics and Astronomy, University of Georgia, USA

² Department of Physics, Kansas State University, USA



**Georgia Physics and Astronomy
Education Research**

Franklin College of Arts and Sciences

UNIVERSITY OF GEORGIA

**KANSAS STATE
UNIVERSITY**

College of Arts & Sciences

Department of Physics



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Discipline-specific postgraduate degree completion is important but difficult to determine across programs

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- Existing data is limited
- Data is often aggregated, which can hide trends between groups of students
- Tracking individual students is resource-intensive when done at scale.

We can estimate these
efficiently with commonly-
collected data.

What information we need



Degrees awarded (US Integrated Postsecondary Education Data System)



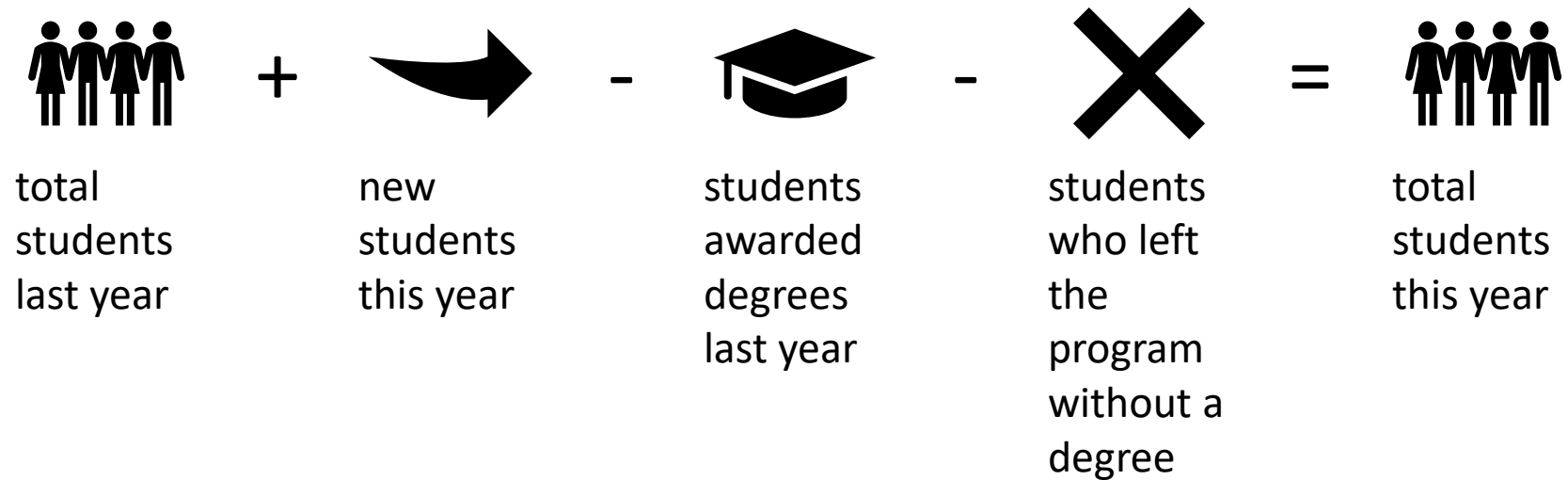
Total number of students (US Survey of Graduate Students and Postdoctorates in Science and Engineering)



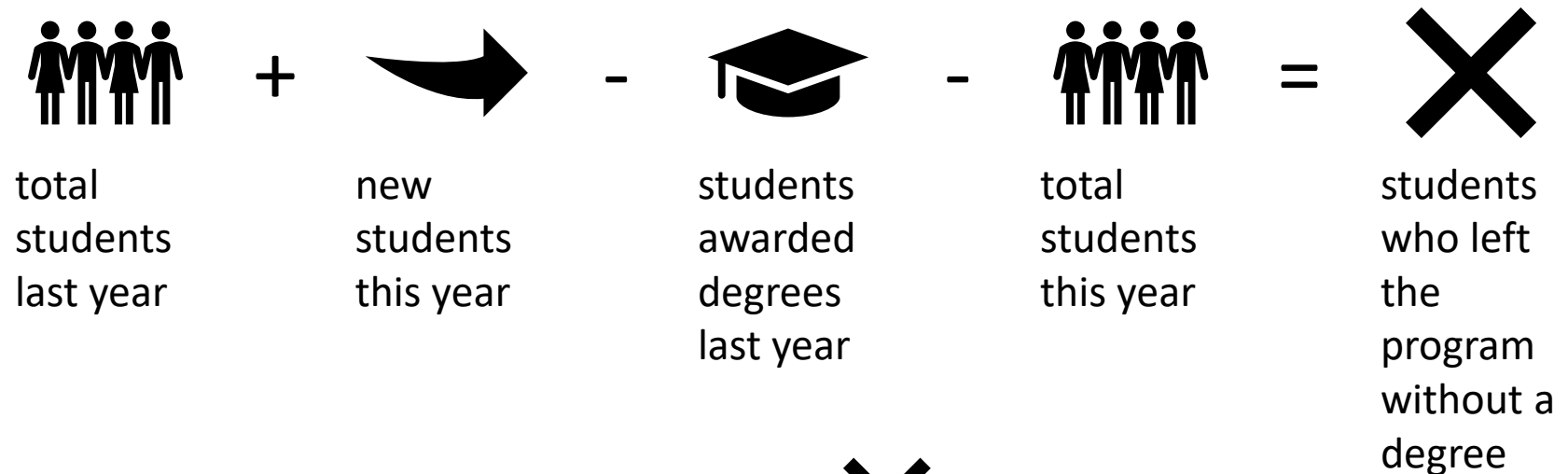
Number of new students (US Survey of Graduate Students and Postdoctorates in Science and Engineering)

Data in today's talk: N= 173 universities

Annual retention rate



Annual retention rate



Retention rate: $1 - \frac{\text{X}}{\text{4 student icons}}$

[Bridges et al. 2024](#)

Completion rate

Multiply N retention rates together to get the program completion rate, where N is the typical program length

$$1 - \frac{X}{\text{people}} * 1 - \frac{X}{\text{people}} * 1 - \frac{X}{\text{people}} * \dots$$

For Physics PhDs* in US, $N \sim 6$ years

*Physics PhDs include ~ 2 years master + 4 years of research

[AIP Physics Trends, 2025](#)

PhD Completion rate

PhD Completion rate: $\frac{\text{Graduation Cap Icon}}{\text{Curved Arrow Icon}}$

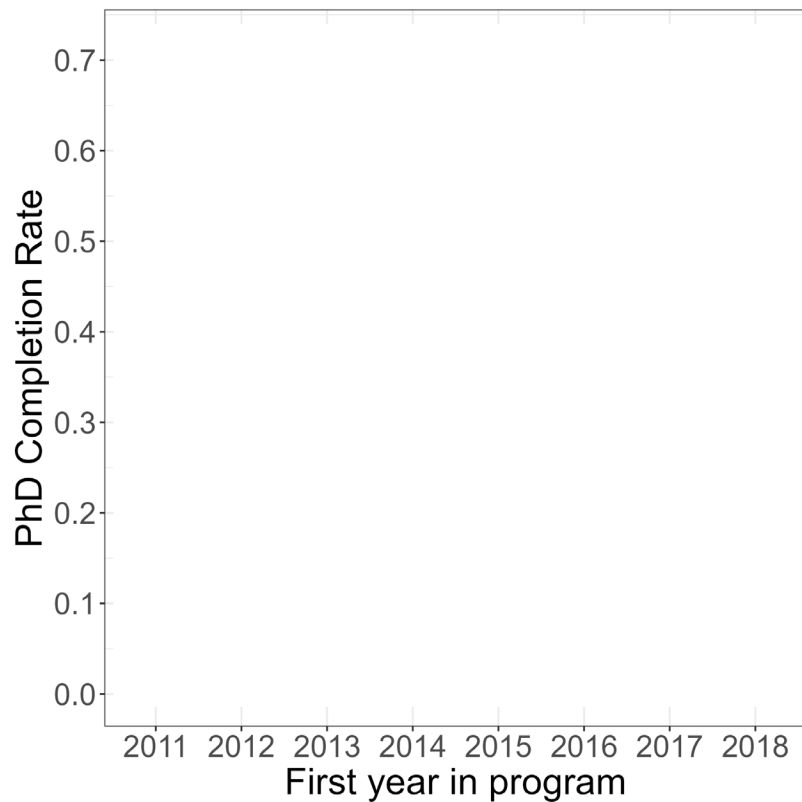
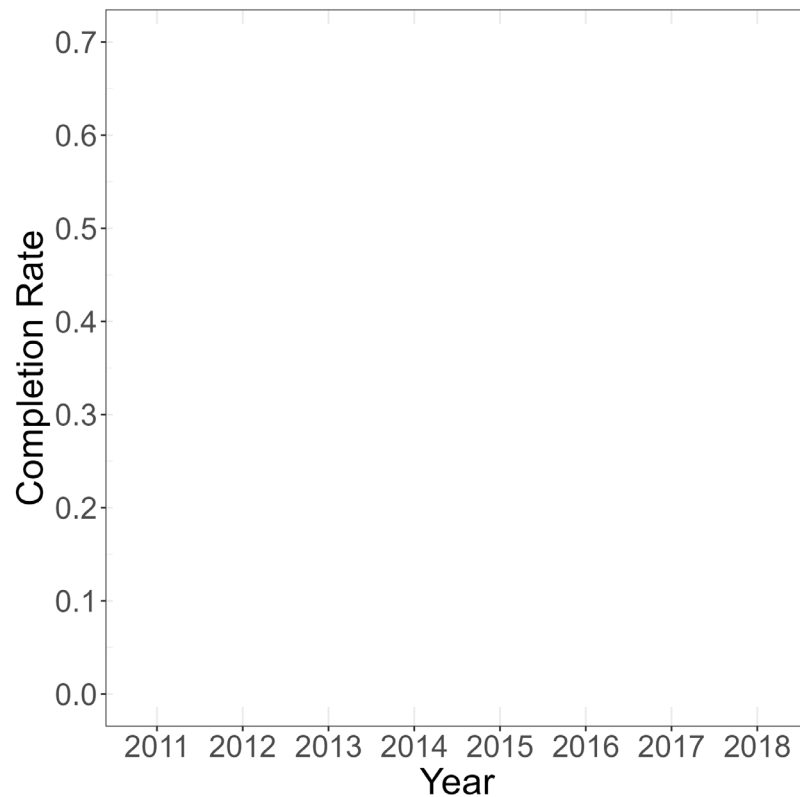
Degrees award N years later*

New students this year*

* average numbers over +/- 1 year for better results

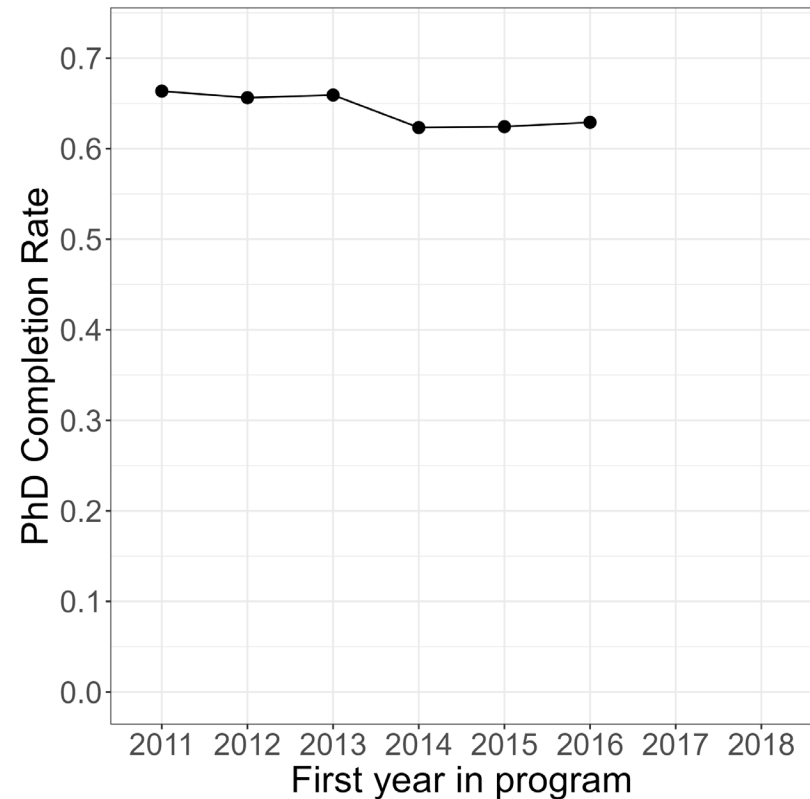
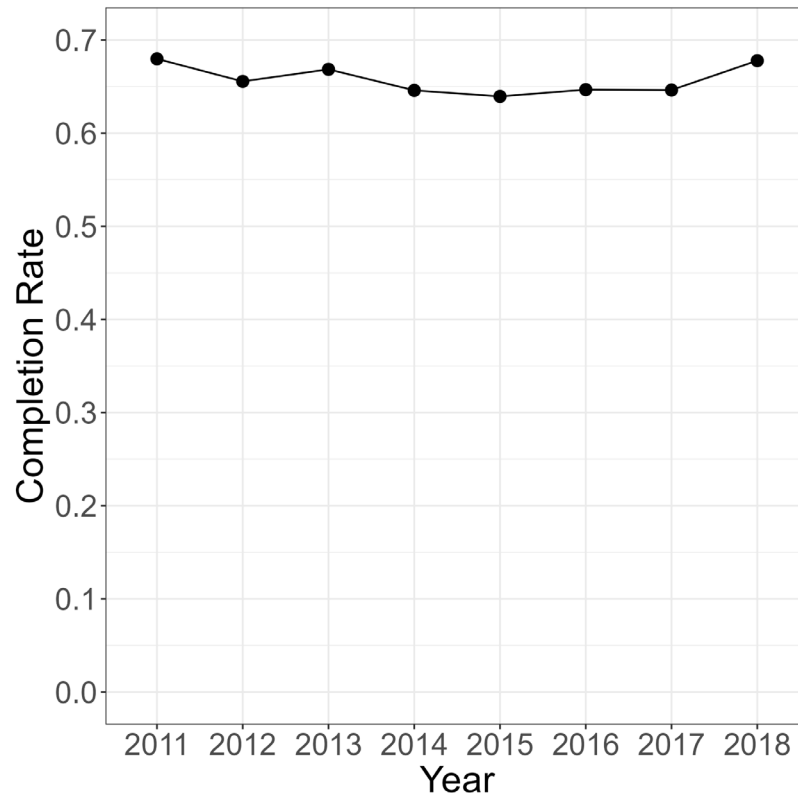
[Bridges et al. 2024](#)

Estimated completion rates in physics



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Estimated completion rates in physics are around 65% and largely constant



Strength of approach: we can investigate specific groups

For each group, we need:



Degrees awarded



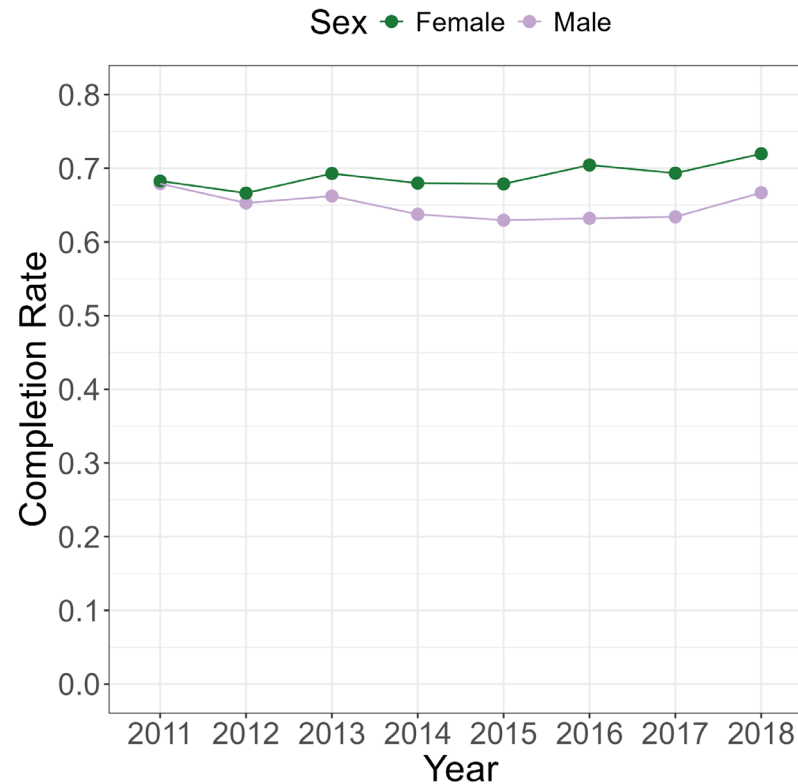
Total number of students



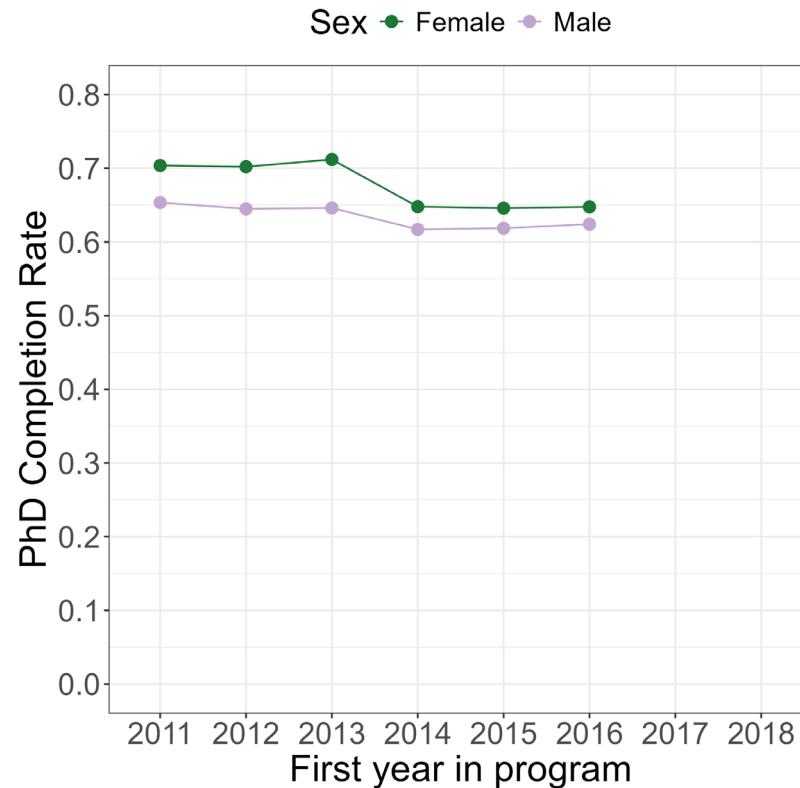
Number of new students

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Women have a slightly higher completion rate than **men** and it largely increases over time

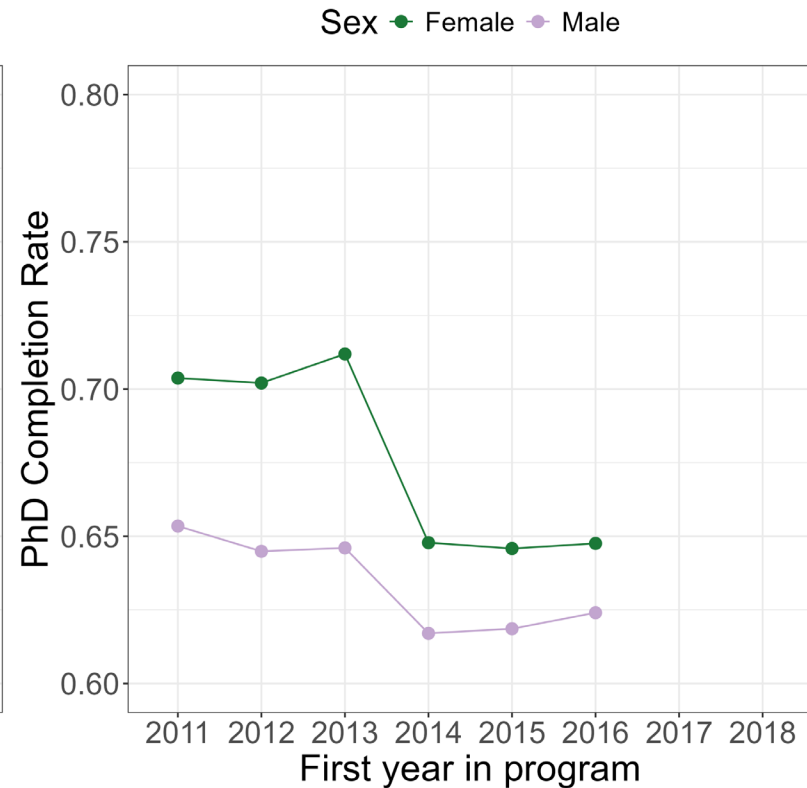
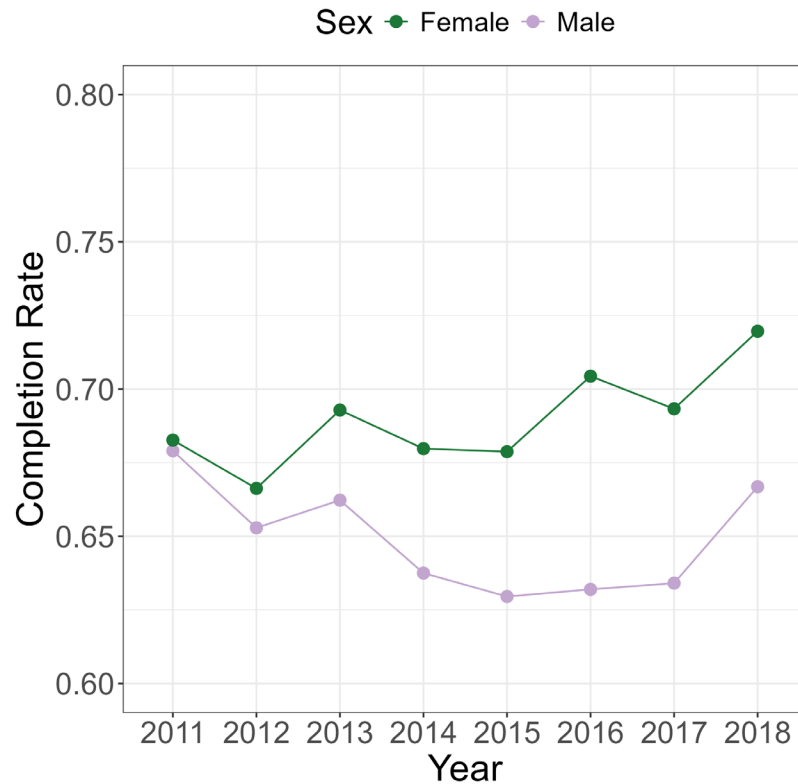


Women have a slightly higher PhD completion rate than men but that rate decreases over time



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Zooming in: similar but different conclusions



What's next: specific groups of students

- race/ethnicity
- sex x race/ethnicity

Longer term: connect completion rates to program practices

- Groupings can also be by program
- Identify programs with higher completion rates and lower completion rates
- Compare program practices to identify potential high-impact ones for PhD completion

Takeaway: With three simple numbers, you too can determine degree completion



Degrees awarded



Total number of students



Number of new students

Takeaway: With three simple numbers (🎓 👥 ➡),
you too can determine degree completion

But there are some caveats:

- data required over many years
- careful consideration of which data to include



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